TECHNICAL TEST RESULTS

1. Test Summary

The attendance system was evaluated through black-box functional testing, boundary-value testing, authentication testing, performance and concurrency testing, and recovery testing. The results below are based on the captured automated test output.

Test Area	Executed	Passed	Failed	Result
QR/attendance scanning	14	12	2	12 of 14 passed
Login/authentication	6	6	0	All passed
Latency measurement	100 samples	100	0	Completed
Concurrency	4 load levels	4	0	Completed
Recovery	4	4	0	Completed

Overall functional result: 18 of 20 functional cases passed (90%).

2. Black-Box Attendance Scan Testing

Each scan input was compared with the expected acceptance decision and, when applicable, the expected attendance status.

No.	Test Case	Expected	Actual	Result
1	Valid QR, enrolled student, 14 minutes after start	True / PRESENT	True / PRESENT	PASS
2	Same QR scanned again immediately	False	False	PASS
3	Unknown QR token	False	False	PASS
4	Expired QR token	False	False	PASS
5	Revoked QR token	False	False	PASS
6	Room QR passed to the student scanner	False	False	PASS
7	Empty QR input	False	False	PASS
8	Valid QR for a student not enrolled in the section	False	False	PASS
9	Boundary: 14 min 59 s after start (inside 15 min grace)	True / PRESENT	True / PRESENT	PASS
10	Boundary: 15 min 01 s after start (grace exceeded)	True / LATE	True / LATE	PASS
11	16 minutes after start	True / LATE	True / LATE	PASS
12	Scan one hour before the scheduled start	False	True / PRESENT	FAIL
13	Repeat scan more than 2 minutes after the first record	False	True / PRESENT	FAIL
14	Scan after the session has ended	False	False	PASS

Result: 12 cases passed and 2 cases failed.

Observed Messages

No.	System Message	Result
1	Test, Student1 marked PRESENT	PASS
2	Already marked as PRESENT less than 2 minutes ago.	PASS
3	Invalid or expired QR code.	PASS
4	Invalid or expired QR code.	PASS
5	Invalid or expired QR code.	PASS
6	This is not a student QR code.	PASS
7	Invalid or expired QR code.	PASS
8	Student is not enrolled in this class section.	PASS

No.	System Message	Result
9	Test, Student2 marked PRESENT	PASS
10	Test, Student3 marked LATE	PASS
11	Test, Student4 marked LATE	PASS
12	Test, Student5 marked PRESENT	FAIL
13	Test, Student1 marked PRESENT	FAIL
14	No active session.	PASS

3. Boundary and Business-Rule Findings

The 15-minute grace boundary behaved as expected: a scan at 14 minutes 59 seconds was marked PRESENT, while a scan at 15 minutes 01 second was marked LATE. Two business-rule cases did not match the expected outcome.

Finding	Expected	Observed	Assessment
Scan one hour before scheduled start	Reject scan	Accepted as PRESENT	FAIL: the active-session rule permitted an early scan.
Repeat scan more than two minutes later	Reject duplicate	Accepted as another PRESENT record	FAIL: duplicate protection only covered the immediate interval.

Recommended action:

- Enforce the allowed scan window against the scheduled session start time.
- Enforce one attendance record per student per session, independent of elapsed time.

4. Attendance Record Integrity

The scan run stored 6 attendance records. Student 1 has 2 records, consistent with the failed duplicate-scan case.

Student	Record Count	Recorded Statuses
1	2	PRESENT,PRESENT
2	1	PRESENT
3	1	LATE
4	1	LATE
5	1	PRESENT

5. Date and Report Validation

Item	Captured Value
Session time (UTC)	2026-10-07 23:00
UTC date	07/10/2026
Local/machine date	08/10/2026
Time zone	Singapore Standard Time
Local-date report	0 sessions; 0 records
UTC-date report	1 session; 6 records; 4 present; 2 late

The same session falls on different UTC and local calendar dates. Report filtering returned data for the UTC date but no data for the local date, indicating a date/time-zone alignment issue that should be reviewed.

6. Login and Authentication Testing

Valid and invalid account states were checked against the expected authentication response.

Test Case	Expected	Actual	Result
Valid username and password	True	True	PASS
Valid username, wrong password	False	False	PASS
Unknown username	False	False	PASS
Locked account	False	False	PASS
Inactive account	False	False	PASS
Account with login disabled	False	False	PASS

Result: All 6 login cases passed.

Authentication Messages

Test Case	System Message
Valid username and password	-
Valid username, wrong password	Invalid username or password. Please try again.
Unknown username	Invalid username or password. Please try again.
Locked account	Your account has been locked. Please contact an administrator.
Inactive account	Your account is inactive. Please contact an administrator.
Account with login disabled	Login is disabled for this account.

7. Performance and Load Testing

Latency was measured over 50 requests for both scanning and report generation.

Operation	Samples	Average	P50	P95	Maximum
Scan	50	5.7 ms	4.4 ms	12.7 ms	29.2 ms
Report	50	4.2 ms	3.8 ms	7.5 ms	9.2 ms

The measured scan and report operations completed with low average and 95th-percentile latency in this test environment.

8. Concurrency Testing

The system was exercised at 50, 100, 200, and 500 concurrent operations.

Users	Successful	Failed	Wall Time	Average	P95	Maximum	Throughput
50	50	0	885 ms	103.0 ms	406.0 ms	639.0 ms	56.5/s
100	100	0	2066 ms	146.4 ms	948.0 ms	2065.0 ms	48.4/s
200	200	0	2763 ms	103.3 ms	636.0 ms	1897.0 ms	72.4/s
500	500	0	6223 ms	186.7 ms	1114.0 ms	2881.0 ms	80.3/s

All concurrency runs completed without failed requests, and the stored-record counts matched the requested load at each level.

Key observations:

- Highest measured throughput: 80.3 operations per second at the 500-user level.
- Highest P95 latency: 1,114 ms at the 500-user level.
- Highest maximum latency: 2,881 ms at the 500-user level.
- No concurrency errors were reported.

9. Recovery Testing

Recovery checks verified record persistence, transaction rollback, and service behavior after restart/release.

Recovery Test	Expected / Action	Observed Result
Restart	3	3 records; recovered
Interrupted Transaction	3	3 records; rolled back cleanly
Resume After Release	Resume data access and scan response	Read: ok; sessions: 1; scan responded: yes; Already marked as PRESENT less than 2 minutes ago.
Final Record Count	3	3

Result: Recovery behavior matched the captured expectations, including clean rollback of the interrupted transaction.

10. Captured Resources

Resource	Captured Value
LogMessages	-
ErrorMessages	Invalid username or password. Please try again.

11. Conclusion

Authentication, latency, concurrency, and recovery tests completed successfully. Functional scanning passed 12 of 14 cases. Before release, priority should be given to preventing scans before the scheduled start, enforcing one attendance record per student per session, and aligning report-date filtering with the intended local time zone.